Rithul Raveendran

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Summary

AI & Machine Learning student with hands-on project experience in Python, backend development, LLM-based applications, and Retrieval-Augmented Generation (RAG). Interested in building practical AI solutions using agentic workflows and intelligent automation.

Education

St. Joseph's College of Engineering 2025 – 2029
B.Tech in Artificial Intelligence & Machine Learning (CGPA: 8.67 / 10.00) *Chennai, Tamil Nadu*

Experience

BitBridge Technologies Jun 2026
AI Engineer Intern *Chennai, India*

- Configured the Luckfox Pico Pro Max (RV1106) embedded board and its software environment to run a YOLO-based object detection model.
- Deployed and executed the model on the board, verifying on-device inference and object detection outputs.

Projects

Aavaaz | *Python, FastAPI, LangGraph, PostgreSQL, pgvector, librosa*

- Developed a multimodal crisis monitoring platform combining voice characteristic analysis using librosa with conversational sentiment assessment.
- Orchestrated AI assessment workflows using LangGraph and implemented a Retrieval-Augmented Generation (RAG) pipeline with PostgreSQL and pgvector to retrieve relevant contextual information for crisis assessment and escalation.

CarbonLens AI | *Python, Flask, Groq, Llama 3.3, Firebase*

- Developed an AI-powered Scope 3 emissions platform using Llama 3.3 to extract activity data from supplier invoices and utility documents in PDF and Excel formats.
- Implemented deterministic emission calculations, role-based supplier and corporate portals, human-in-the-loop verification, and automated BRSR PDF reporting.

Mrithasanjeevani | *Python, Flask, SQLite, SpeechRecognition, WebSockets*

- Built a healthcare coordination platform using Flask and SQLite, with role-based access for patient, doctor, and pharmacy workflows.
- Implemented real-time emergency SOS alerts using WebSockets and Geolocation APIs, alongside multilingual voice-based symptom recording using SpeechRecognition.

Technical Skills

Languages: Python, C, SQL
Frameworks: FastAPI, Flask, React
AI/ML Libraries & Technologies: LangGraph, NumPy, Pandas, librosa, SpeechRecognition, YOLOv5
Databases: PostgreSQL, SQLite, Firebase, Supabase
Tools & Technologies: Linux, Git, GitHub, WebSockets, REST APIs, RKNN Toolkit2
Core Competencies: LLM Integration, Retrieval-Augmented Generation (RAG), AI Workflow Orchestration, Backend Development, On-Device AI Inference

Achievements

- **Winning Team** – Makeathon 2.0
- **Winning Team** – Makeathon 3.0

Certifications

- **Oracle:** Agentic AI Certified Foundations Associate
- **AWS Academy Graduate:** Machine Learning for NLP, Cloud Foundations, Cloud Operations
- **Professional Credentials:** Python for Data Science (NPTEL), Introduction to AI & ML (Infosys), Networking Essentials (Cisco)